

1. General Description

The EMS3412 is a high-speed bidirectional passive switch in mux or demux configurations suited for USB Type-C™ application supporting USB2.0, USB 3.0 data rates. Based on control pin SELx, the device provides switching on differential channels between Port A, or B to Common Port. The EMS3412 is a generic analog differential passive switch that can work for any high-speed interface applications requiring a common mode voltage range of 0 to 2 V and differential signaling with differential amplitude up to 1.8Vpp. It employs adaptive tracking that ensures the channel remains unchanged for the entire common mode voltage range. Excellent dynamic characteristics of the device allow high-speed switching with minimum attenuation to the signal eye diagram with very little added jitter. The EMS3412 is available in QFN 4x4-24L with Pb-free and Halogen-free making it a perfect candidate for mobile and space constrained applications.

2. Features and Benefits

- Two Differential Channel 1:2/ 2:1 Mux/ De-Mux
- USB 2.0/ 3.0/ HDMI 2.1/ SATA/ DP compatible
- USB 3.0 Super Speed 10 Gbps Switch, Bandwidth: 5.1 GHz @ -3 dB insertion loss
- Supports both AC coupled and DC coupled signals
- Isolation: -27 dB @ 1.2 Gbps
- Crosstalk: -43 dB @ 1.2 Gbps
- ESD Tolerance: 2 kV HBM
- Low bit-to-bit skew, Bidirectional
- Wide V_{CC} Operating Range: 1.5 V to 5.0 V
- Small Packaging, QFN 4 x 4-24 Lead

3.Applications

- USB Type-C Ecosystem
- Desktop and Notebook PCs
- Server/Storage Area Networks
- PCI Express Backplanes
- Shared I/O Ports
- FPD LinkII and FPD LinkIII Switching

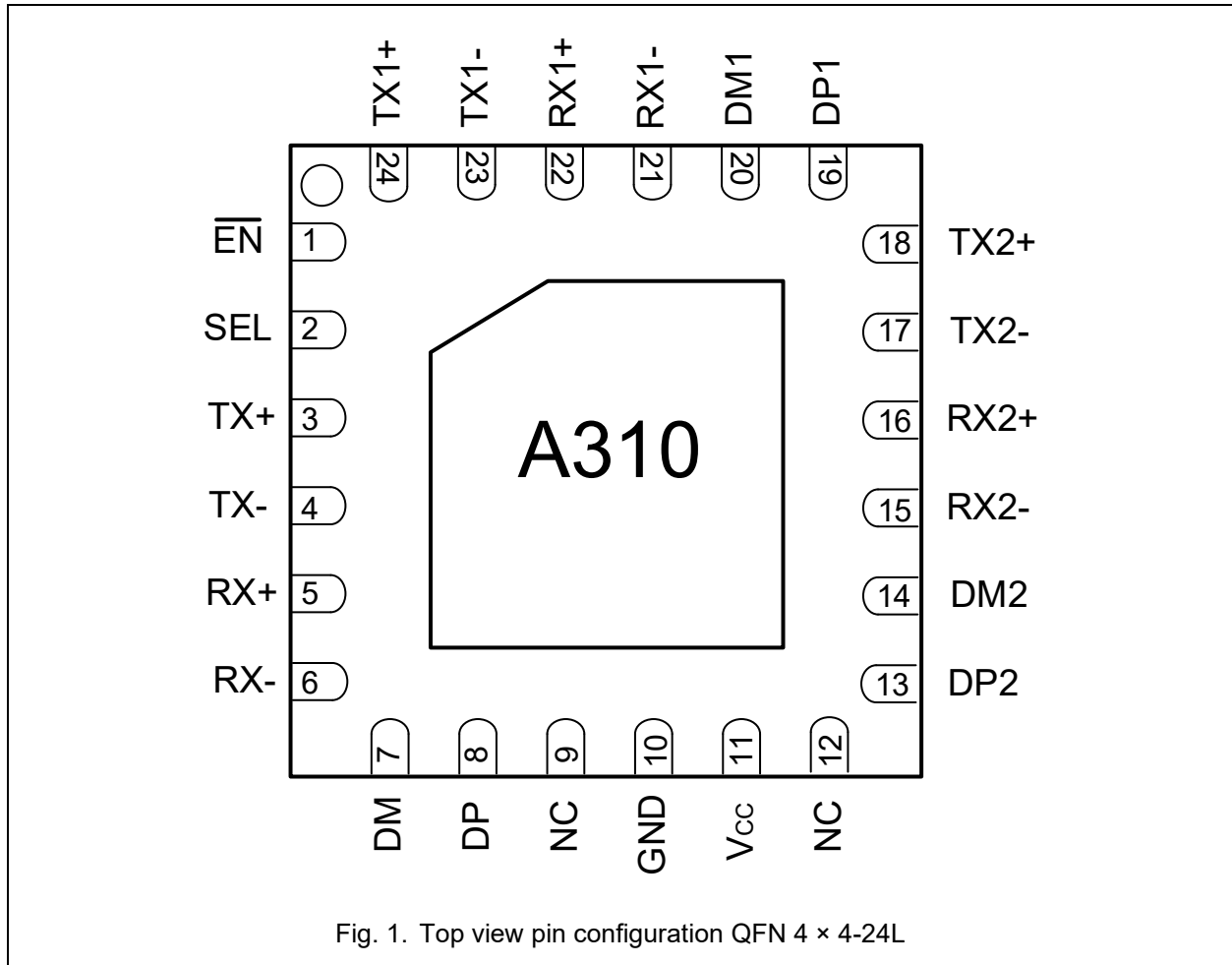
4.Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3412	A310	QFN 4 × 4-24L	Quad Flat No-leads Package, 24 pins; body width 4 mm; 0.5 mm pitch, 0.75 mm height	3000

5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Signal Type	Description
$\overline{\text{EN}}$	1	I	Chip Enable, Active Low
SEL	2	I	Switch logic control
TX+	3	I/O	Super Speed TX+ Common
TX-	4	I/O	Super Speed TX- Common
RX+	5	I/O	Super Speed RX+ Common

EMS3412

2:1 Mux/De-Mux USB2.0 + USB3.0 Compliant Signal Switch

RX-	6	I/O	Super Speed RX- Common
DM	7	I/O	USB 2.0 Common Port
DP	8	I/O	USB 2.0 Common Port
NC	9		Not Connected
GND	10	Ground	Power Ground
V _{CC}	11	Power	Positive Supply Voltage
NC	12		Not Connected
DP2	13	I/O	USB 2.0 Common Port B
DM2	14	I/O	USB 2.0 Common Port B
RX2-	15	I/O	Super Speed RX2-
RX2+	16	I/O	Super Speed RX2+
TX2-	17	I/O	Super Speed TX2-
TX2+	18	I/O	Super Speed TX2+
DP1	19	I/O	USB 2.0 Common Port A
DM1	20	I/O	USB 2.0 Common Port A
RX1-	21	I/O	Super Speed RX1-
RX1+	22	I/O	Super Speed RX1+
TX1-	23	I/O	Super Speed TX1-
TX1+	24	I/O	Super Speed TX1+

6.Functional Description

Table 3. Function table

Hi-Z = HIGH Impedance; X = don't care.

$\overline{\text{EN}}$	SEL	TX+	TX-	RX+	RX-	DM	DP
High	X	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z	Hi-Z
Low	Low	TX1+	TX1-	RX1+	RX1-	DM1	DP1
Low	High	TX2+	TX2-	RX2+	RX2-	DM2	DP2

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Min	Max	Unit
V _{CC}	Supply Voltage to Ground Potential	-0.5	5.5	V
V _{TX/RX}	Super Speed Data Channel TX/RX	-0.5	3.8	V
V _I	DC Input Voltage	-0.5	V _{CC}	V
I _O	DC Output Current		50	mA
T _{stg}	Storage Temperature	-65	150	°C
T _j	Junction Temperature		125	°C
P _D	Power Dissipation		300	mW

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Min	Max	Unit
V _{CC}	supply voltage	1.5	5	V
V _I	control input voltage (SEL, $\overline{\text{EN}}$)	0	5	V
V _{SW}	high speed switch I/O voltage	0	3.3	V
T _{amb}	ambient temperature	-40	85	°C

9. Electrical Characteristics

Table 6. Electrical characteristics (Ta = 25 °C, V_{CC} = 1.8 V, unless otherwise specified.)

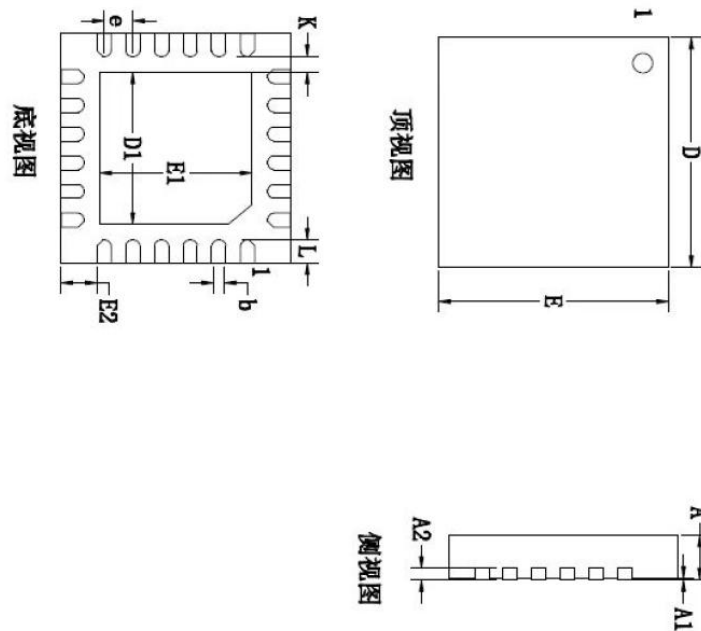
Symbol	Parameter	Conditions	Min.	Typ.	Max.	UNIT
POWER SUPPLY						
V _{CC}	supply voltage		1.5		5	V
I _Q	V _{CC} Quiescent Current	SEL = 0 or V _{CC} , $\overline{\text{EN}} = 0$		28		V
I _{PD}	Power-down Current	SEL = 0 or V _{CC} , $\overline{\text{EN}} = \text{V}_{\text{CC}}$			1	μA
DC CHARACTERISTICS						
V _{IH}	control input logic high	V _{CC} = 1.8 to 4.5 V	1.6			V
V _{IL}	control input logic low	V _{CC} = 1.8 to 4.5 V			0.4	V
R _{UP}	$\overline{\text{EN}}$ internal pull-up resistor			2		MΩ
R _{DN}	SEL internal pull-down resistor			2		MΩ
R _{ON_HS}	On-Resistance for TX/RX	V _{IS} = 0.2 V, I _{ON} = 8 mA		6.7	8	Ω
R _{FLAT_LP}	R _{ON} Flatness for TX/RX [1]	V _{IS} = 0 to 1.2 V, I _{ON} = 8 mA		0.8	1	Ω
R _{FLAT_LP}	R _{ON} Flatness for TX/RX	V _{IS} = 0 to 0.2 V, I _{ON} = 8 mA		0.2	0.3	Ω
R _{MATCH}	R _{ON} Matching Between Channels [2]	V _{IS} = 0 to 1.2 V, I _{ON} = 8 mA		0.1		Ω
I _{OFF}	Switch Off Leakage Current	$\overline{\text{EN}} = \text{V}_{\text{CC}}$, Tx, Rx = V _{CC} , TX1, TX2, RX1, RX2 = 0	-0.5		0.5	μA
AC CHARACTERISTICS						
t _{EN}	Enable Time $\overline{\text{EN}}$ to Output	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		80	150	μs
t _{DIS}	Disable Time $\overline{\text{EN}}$ to Output	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		40	250	ns
t _{ON}	Turn-On Time SEL to Output	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		400	1200	ns
t _{OFF}	Turn-Off Time SEL to Output	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		130	800	ns
t _{BBM}	Break-Before-Make Time	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		250	500	ns
t _{PD}	Propagation Delay	R _L = 50 Ω, C _L = 0 Pf, V _{IS} = 0.6V		0.25		ns
Off	Off Isolation	R _L = 50 Ω, f = 1.2 GHz, V _{IS} = 0.2V _{PP}		-27		dB
X _{TALK}	Crosstalk [3]	R _L = 50 Ω, f = 1.2 GHz, V _{IS} = 0.2V _{PP}		-43		dB
BW _{-3dB}	-3 dB Bandwidth	R _L = 50 Ω, C _L = 0 pF, Signal 0 dBm	4.5	5.1		GHz
CAPACITANCE						
C _{ON}	Switch On Capacitance	V _{Bias} = 0.2 V, f = 1.5 GHz		1.5		pF
C _{OFF}	Switch Off Capacitance	V _{Bias} = 0.2 V, f = 1.5 GHz		1.0		pF

Note:

- [1] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.
 [2] R_{ON} matching between channels is calculated by subtracting the channel with the lowest max R_{ON} value from the channel with the highest max R_{ON} value.
 [3] Crosstalk is inversely proportional to source impedance.

10. Package Outline

QFN 4 × 4-24L



SYMBOL	MILLMETER		
	MIN	NOM	MAX
A	0.70	0.75	0.80
A1	0.00	—	0.05
A2	0.203 TIY		
b	0.20	0.25	0.30
D	3.95	4.00	4.05
D1	2.55	2.65	2.75
E	3.95	4.00	4.05
E1	2.55	2.65	2.75
E2	0.625 TIY		
e	0.50 BSC		
K	0.275 BSC		
L	0.35	0.40	0.45

11. Abbreviations

Table 7. Abbreviations

Acronym	Description
ESD	ElectroStatic Discharge
HBM	Human Body Model

12. Revision History

Table 8. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3412 Rev. 0.1	Sep 28, 2025	Draft datasheet		